probability distribution. Let ϕ be a C^∞ function of the phase variables with compact support and define the time t average by

$$\bar{\phi}(t) = \frac{\int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega}{\int_{\mathcal{D}} \rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega}. \tag{5.6}$$

As the flow is Hamiltonian, $\int_{\mathcal{D}} \rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega \equiv \int_{\mathcal{D}} \rho(\boldsymbol{q},\boldsymbol{p},0)\mathrm{d}\omega$. Thus the condition for $\bar{\phi}$ to converge is just that

$$\int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\frac{\partial\rho}{\partial t}(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega \to 0.$$

Since

$$\frac{\partial\rho}{\partial t} = \mathcal{L}_f^\dagger \rho,$$

this suggests that the condition for the invariant probability density function ρ_∞ should be

$$\int_{\mathcal{D}} \phi \mathcal{L}_f^\dagger \rho_\infty \mathrm{d}\omega = 0,$$

for all suitable functions $\phi \in C_0^\infty(\mathbb{R}^m)$. We say that ρ_∞ satisfies

$$\mathcal{L}^\dagger \rho_\infty = 0,$$

in the *weak* or *variational* sense. When such a density exists, we refer to the associated probability distribution as an *invariant distribution*. Such invariant distributions are essential tools in molecular modelling. In the case of a Hamiltonian system, we know that $\mathcal{L}_f^\dagger = -\mathcal{L}_f$, and, moreover,

$$\mathcal{L}_f H = 0,$$

so that H is a first integral. Any probability distribution whose density is a smooth function of H will also be an invariant distribution.

In particular, consider the density

$$\rho_\varepsilon(\boldsymbol{q},\boldsymbol{p}) = Z_\varepsilon^{-1} e^{-(H(\boldsymbol{q},\boldsymbol{p})-\mathrm{E})^2/2\varepsilon},$$

where ε is an arbitrary small positive constant and E is an accessible energy level. We assume that H tends to infinity sufficiently rapidly as $\|\boldsymbol{z}\| \to \infty$ so that the exponential decays sufficiently rapidly (and thus the total integral exists) and take